

		$a_1 : a_2 = 1.0 : 0.2 = 5 : 1$
5	B	The board is in equilibrium. Net moment about point P is zero. Let x be the perpendicular distance between the line of action of force $2.5F$ at R to P. $F(20) + 0.5F \sin 45^\circ (x) = 2.5F(x) + 0.5F \cos 45^\circ (20)$ $2.146(x) = 12.93$ $x = 6.0 \text{ cm}$ Therefore, the area of the board = $(40)(12) = 480 \text{ cm}^2$
6	D	After the stone leaves the hand of the astronaut, it does not experience any